

Bacterial Genetics & Virulence Determinants

Current 48-slide faculty deck primary - official Week 4 objectives integrated - horizontal transfer linked directly to resistance and toxin acquisition

3 official objectives

Transformation

Conjugation

Transduction

Resistance genes

Lysogenic conversion

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Describe the mechanisms of genetic exchange in bacteria.
2. Explain how antibiotic resistance can be transmitted among bacteria.
3. Describe how bacteriophages can spread virulence factors and contribute to the evolution of bacterial pathogens.

ASSIGNED SCHOLARRX

Bacterial Genetics & Virulence Factors

Vertical transfer passes genes from parent cell to daughter cells. **Horizontal gene transfer** moves DNA between cells and can rapidly spread traits such as resistance or virulence across a bacterial population.

Deck mnemonic: transformation = *USB stick* (free DNA), conjugation = *direct cable* (cell contact), transduction = *virus email* (bacteriophage vector).

Mechanism	DNA source/vector	Cell contact?	DNase sensitive?	Typical cargo
Transformation	Naked/free DNA	No	Yes	Usually smaller DNA fragments/plasmid DNA
Conjugation	Donor cell; plasmid/ chromosomal DNA	Yes	No	Large DNA pieces, often whole plasmids
Transduction	Bacteriophage	No	No - DNA protected in phage	Small bacterial DNA segments / phage-associated genes

Slides 11-16

Transformation: naked DNA uptake

LO 1

Donor DNA released

->

competent recipient takes up DNA

->

recombines into chromosome or persists as plasmid

Diagnostic clue: extracellular DNA is exposed, so DNase blocks transformation. The deck contrasts this with conjugation and transduction, which remain DNase resistant.

Historical logic: Griffith demonstrated a transferable transforming principle; Avery-MacLeod-McCarty identified DNA as that principle; Hershey-Chase independently showed that phage DNA, not protein, enters bacteria.

Gram-negative classic

F plasmid promotes direct contact through the sex pilus; DNA is copied from donor to recipient.

Hfr

When the F factor integrates into the bacterial chromosome, conjugation can drag adjacent chromosomal genes into the recipient.

Gram-positive twist: the deck emphasizes *Enterococcus faecalis*, where peptide pheromones induce donor adhesins and cell clumping instead of a classic sex pilus.

Conjugation clues: requires contact, is DNase resistant, and is especially effective at moving large plasmids carrying multiple resistance determinants.

A bacteriophage packages bacterial DNA or carries its own accessory genes, then delivers DNA into another bacterium. Bacterial cells do not need to touch.

Form	Mechanism	Cargo pattern
Generalized transduction	Accidental packaging of host DNA during phage production	Potentially random bacterial genes
Specialized transduction	Imprecise excision of a temperate prophage	Genes adjacent to the prophage insertion site

Lytic phage

Replicates, assembles progeny, and lyses the host cell.

Temperate phage

Can integrate into the bacterial chromosome as a **prophage**; induction can later return it to a lytic cycle.

DNase does not stop transduction: the transferred DNA is physically protected within the phage particle.

Slides 29-32

Horizontal transfer accelerates antibiotic resistance

LO 2

Resistance gene appears in one lineage

->

plasmid/mobile element captures it

->

horizontal transfer

->

rapid population spread

Deck anchors: MRSA carries *mecA*; vancomycin-resistant enterococci can carry *vanA*. Clinically important resistance often spreads by acquired mobile DNA rather than by waiting for each lineage to evolve the same mutation independently.

Slides 33-40

Lysogenic conversion turns phage genes into virulence factors

LO 3

Lysogenic conversion occurs when a temperate phage integrates as a prophage and a phage-encoded gene changes the bacterial phenotype - classically by adding toxin production.

Phage-borne toxin	Bacterium	Disease anchor from deck
Diphtheria toxin	<i>Corynebacterium diphtheriae</i>	Diphtheria
Cholera toxin	<i>Vibrio cholerae</i>	Cholera
Shiga toxin	STEC / <i>E. coli</i> O157:H7	Hemorrhagic colitis - HUS
Botulinum toxin C/D	<i>Clostridium botulinum</i>	Botulism
Erythrogenic toxin	<i>Streptococcus pyogenes</i>	Scarlet fever

Opening case answer: the deck frames O157:H7 as a lineage made more dangerous when a temperate Shiga-toxin phage lysogenized an ancestral *E. coli* strain.

Suspected EHEC/STEC: the lecture emphasizes supportive care and HUS monitoring rather than reflex antibiotic therapy, because prophage induction can increase Shiga-toxin output. Antimotility agents are also avoided in the deck's clinical framing.

Phage therapy

The deck closes by contrasting phages as agents of horizontal gene transfer with their potential therapeutic use against difficult bacterial infections. The clinical-development point is secondary to the exam-relevant transfer/virulence mechanisms.

Exam traps

Identify the route from the vignette clue

All LOs

Trap 1: DNase sensitivity strongly points to transformation.

Trap 2: A pilus or direct donor-recipient contact points to conjugation.

Trap 3: A bacteriophage vector points to transduction; a prophage toxin points to lysogenic conversion.

Trap 4: Hfr means the F factor is integrated into the bacterial chromosome.

Trap 5: Generalized = random bacterial DNA; specialized = genes near the prophage site.

30-SECOND FINISH

Transformation = naked DNA/DNase sensitive. Conjugation = contact/plasmids/DNase resistant. Transduction = phage/DNase resistant. These same routes spread resistance. Temperate phages can also add toxin genes by lysogenic conversion.